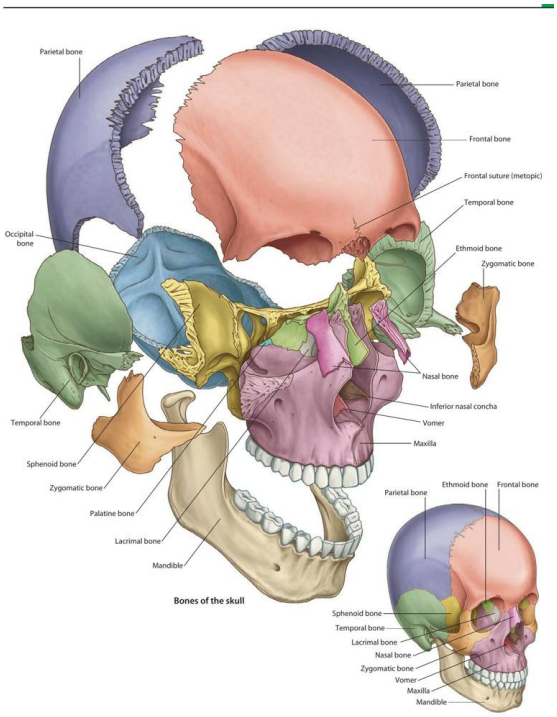
Skull bones

The dry skull for the practical: the views you are handed, the three cranial fossae, every foramen with its contents, and the individual bones.

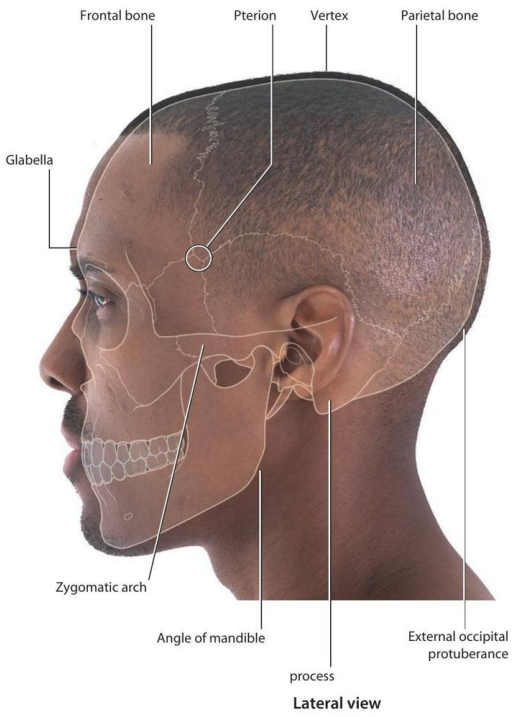
The 22 bones of the skull

- **Cranial bones (8):** frontal, ethmoid, sphenoid and occipital (single); parietal and temporal (paired).
- **Facial bones (14):** nasal, lacrimal, zygomatic, maxilla, palatine and inferior concha (paired); vomer and mandible (single).
- The **ear ossicles** (six in all) are usually counted separately. The **hyoid** is not part of the skull.
- Most skull bones meet at **sutures**. The **temporomandibular joint** is the skull's main synovial joint.

Clinical pearl: If you are asked how many bones are in the skull, state whether you are counting the ossicles.



Bones of the skull (exploded view). Gray's Atlas of Anatomy, 2nd ed., p. 477

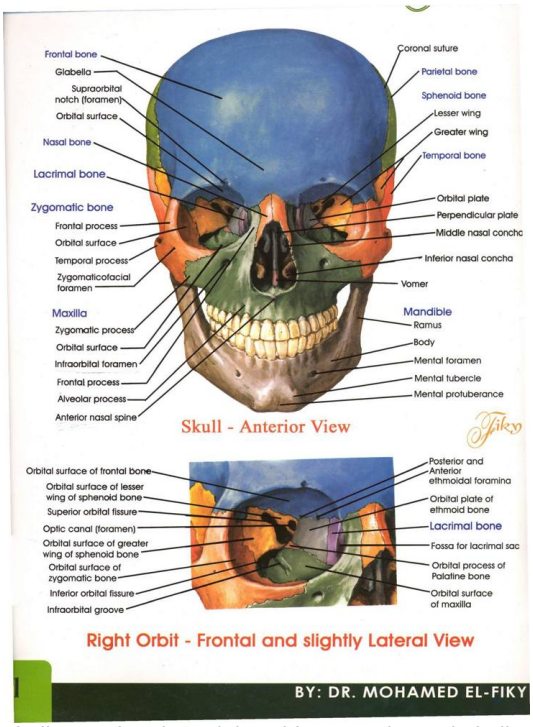


Surface anatomy with bones: lateral view (pterion, mastoid, angle). Gray's Atlas of Anatomy, 2nd ed., p. 476

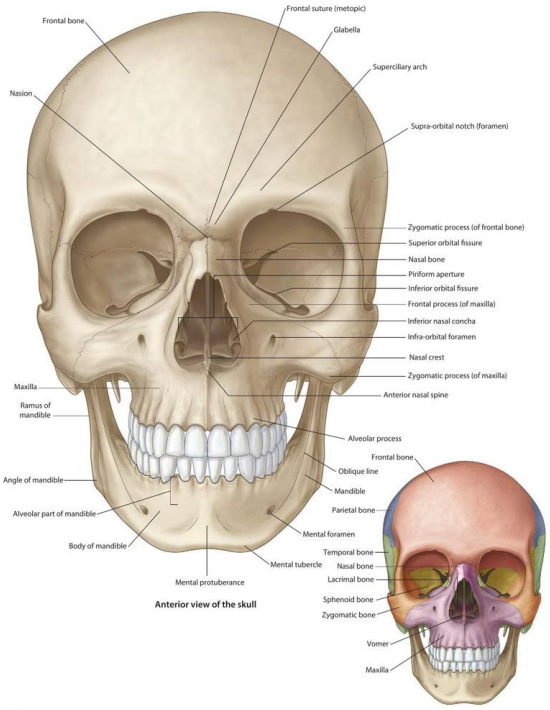
Norma frontalis: the skull from in front

- **Landmarks:** glabella, nasion, superciliary arch, supra-orbital margin, piriform aperture, anterior nasal spine.
- **Three foramina on the face, roughly in one vertical line: supra-orbital** (V1), **infra-orbital** (V2), **mental** (V3).
- **Orbital openings:** optic canal, superior and inferior orbital fissures.
- The **piriform aperture** is bounded by the nasal bones above and the maxillae at the sides and below.

Clinical pearl: The three facial foramina are where you can block each division of the trigeminal nerve.



Skull, anterior view; right orbit. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 1

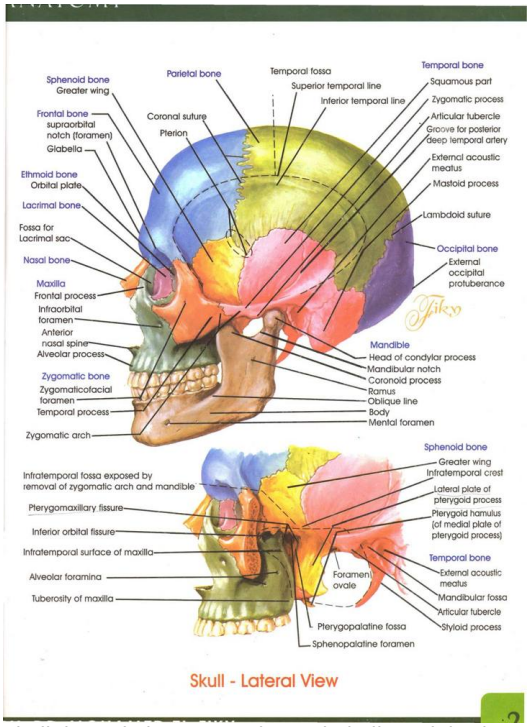


Skull, anterior view: orbits, piriform aperture and maxilla. Gray's Atlas of Anatomy, 2nd ed., p. 478

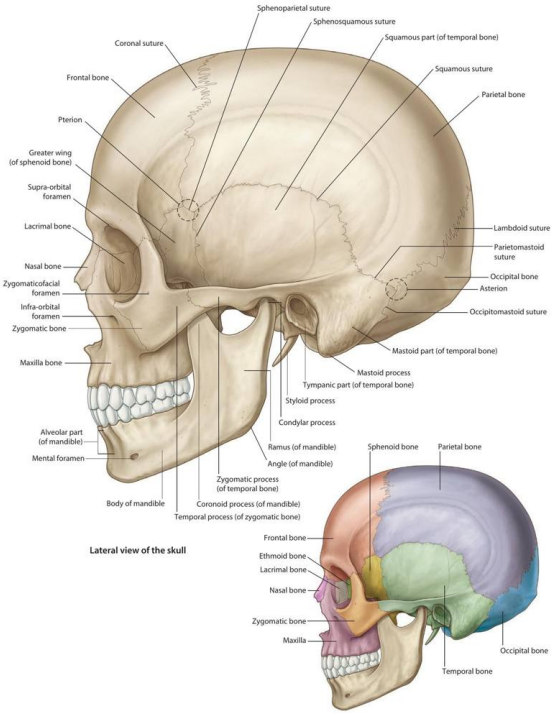
Norma lateralis: the skull from the side

- **Pterion:** an H-shaped junction of **four bones**: frontal, parietal, squamous temporal and greater wing of sphenoid. It lies about 4 cm above the zygomatic arch and about 3 cm behind the frontozygomatic suture.
- **Asterion:** where the **lambdoid, parietomastoid and occipitomastoid sutures** meet. It lies close to the junction of the transverse and sigmoid sinuses.
- **Zygomatic arch:** temporal process of the zygomatic bone plus zygomatic process of the temporal bone.
- **Also identify:** temporal lines, external acoustic meatus, mastoid and styloid processes, mandibular fossa and articular tubercle.

Clinical pearl: Pterion overlies the anterior division of the middle meningeal artery, the classic source of an extradural haematoma.



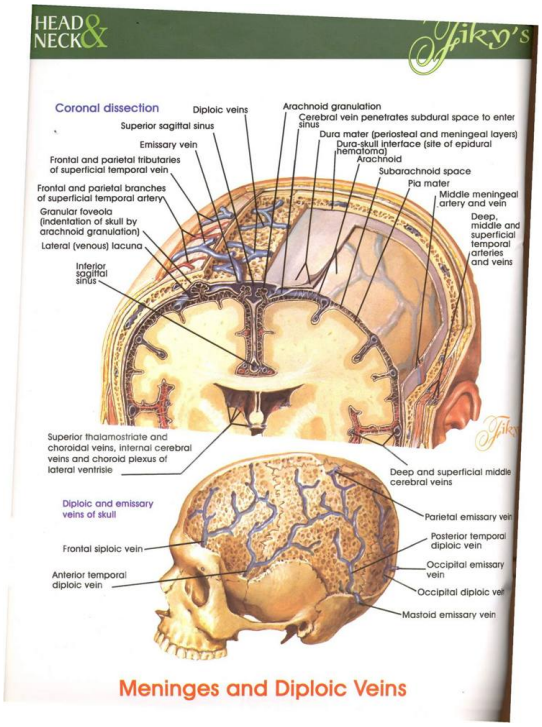
Skull, lateral view. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 2



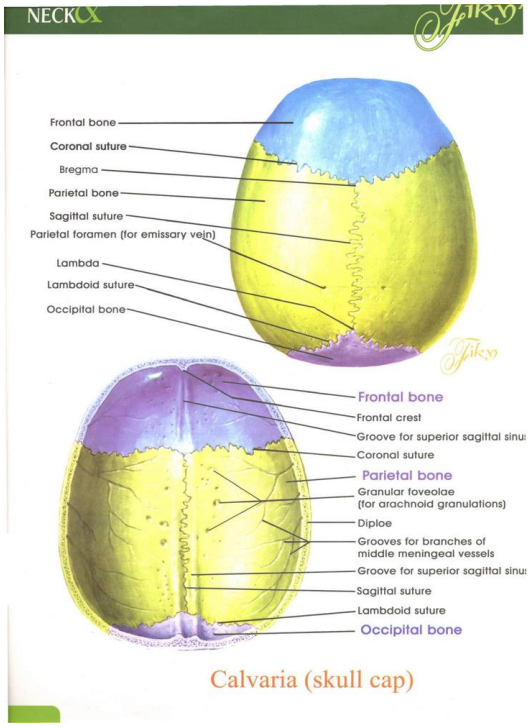
Lateral view of the skull: squamous, tympanic and mastoid parts. Gray's Atlas of Anatomy, 2nd ed., p. 480

Norma occipitalis and verticalis

- **From behind:** lambdoid suture, external occipital protuberance (its midpoint is the inion), superior and inferior nuchal lines, external occipital crest, mastoid process and mastoid notch.
- **From above:** frontal, two parietals and occipital. The sutures are coronal, sagittal and lambdoid.
- **Bregma:** coronal meets sagittal, the site of the anterior fontanelle. **Lambda:** sagittal meets lambdoid, the site of the posterior fontanelle.
- **Sutural (Wormian) bones** are inconstant extra bones, most often in the lambdoid suture.



Meninges and diploic veins. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 47

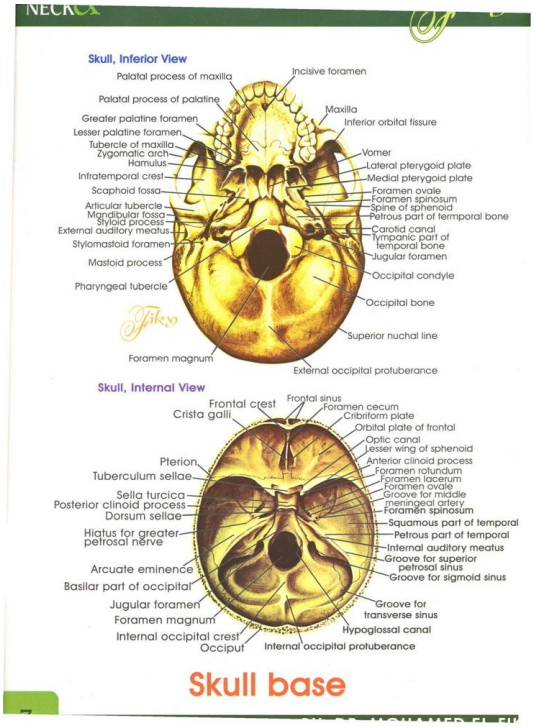


Calvaria (skull cap). Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 3

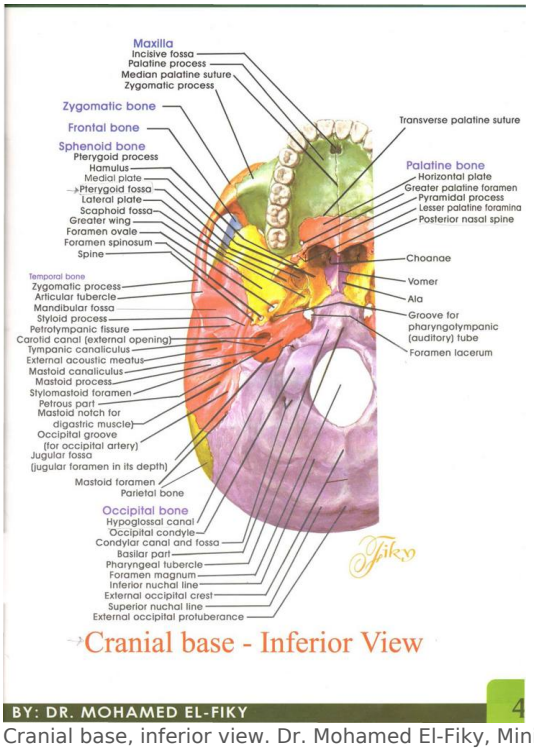
Norma basalis: the skull from below

- **Hard palate:** palatine process of the maxilla in front, horizontal plate of the palatine bone behind.
- **Palatal openings:** incisive fossa in the midline; greater palatine foramen near the third molar; lesser palatine foramina behind it.
- **Middle part:** medial and lateral pterygoid plates, hamulus, scaphoid fossa, foramen ovale, foramen spinosum, foramen lacerum, spine of sphenoid, groove for the auditory tube.
- **Posterior part:** carotid canal, jugular foramen, styloid process, stylomastoid foramen, mastoid process and notch, occipital condyle, hypoglossal canal, foramen magnum, pharyngeal tubercle.

Clinical pearl: The stylomastoid foramen lies between the styloid and mastoid processes. It is where the facial nerve leaves the skull.



Skull base, inferior and interior views. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 7

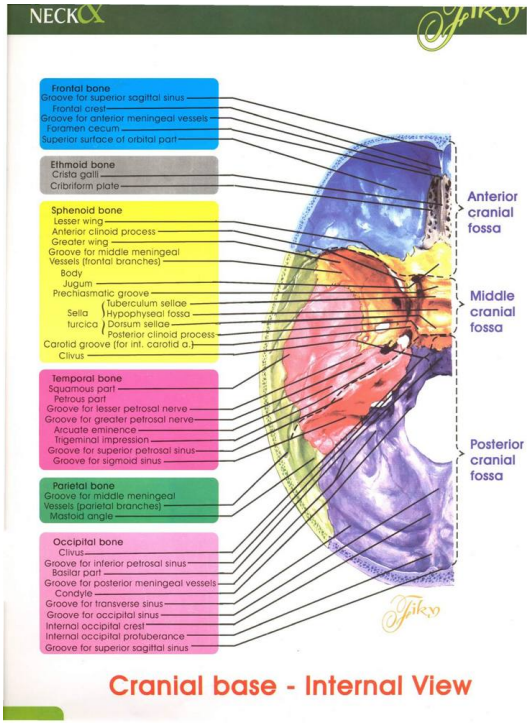


Cranial base, inferior view. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 4

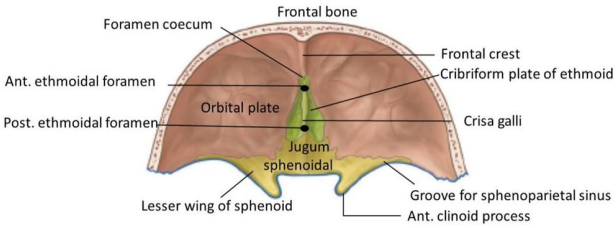
Anterior cranial fossa

- **Floor:** orbital part of the frontal bone, cribriform plate of the ethmoid, and the lesser wings and front of the body of the sphenoid.
- **Crista galli:** gives attachment to the falx cerebri.
- **Cribriform plate:** olfactory nerve filaments. The **anterior ethmoidal nerve and vessels** also enter the nasal cavity near its front.
- **Foramen caecum:** in front of the crista galli; may transmit a small emissary vein.

Clinical pearl: A fracture of the cribriform plate can cause CSF rhinorrhoea and anosmia.



Cranial base, internal view. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 5

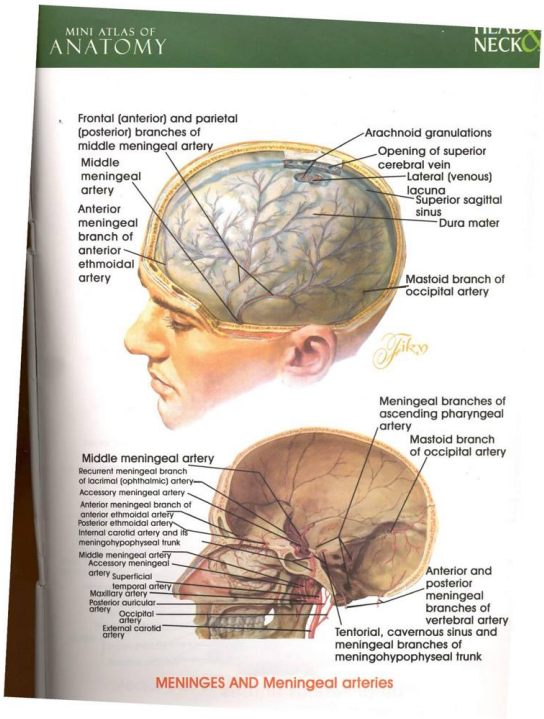


Anterior cranial fossa. Dr. Sameh Doss handout, p. 24

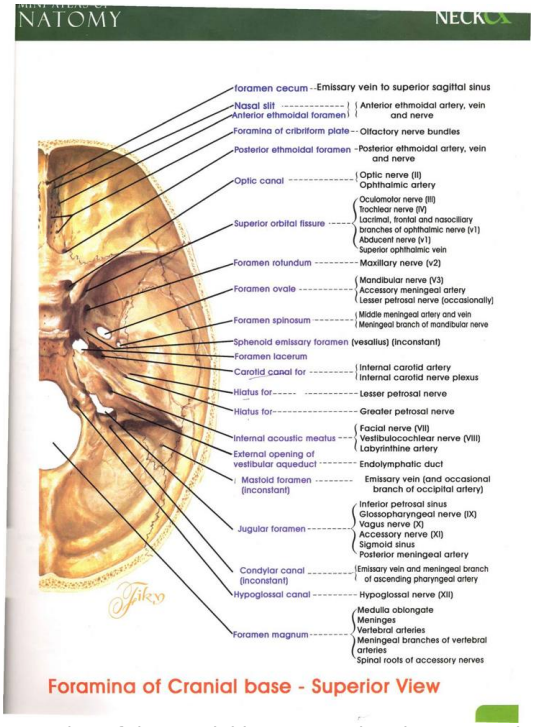
Middle cranial fossa

- **Floor:** greater wing and body of the sphenoid, plus the squamous and petrous parts of the temporal bone. The **sella turcica** lies in the centre.
- **Optic canal:** optic nerve and ophthalmic artery. **Superior orbital fissure:** III, IV, VI, lacrimal, frontal and nasociliary nerves, and the superior ophthalmic vein.
- **Foramen rotundum:** V2. **Foramen ovale:** V3, accessory meningeal artery, lesser petrosal nerve, emissary vein. **Foramen spinosum:** middle meningeal artery and the meningeal branch of V3.
- **Foramen lacerum:** closed by cartilage in life. The internal carotid artery crosses its upper part.
- **Petrous landmarks:** trigeminal impression, hiatus for the greater and lesser petrosal nerves, arcuate eminence, tegmen tympani.

Clinical pearl: The tegmen tympani is a thin plate of bone between the middle ear and the middle cranial fossa, so infection can spread through it.



Meninges and meningeal arteries. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 48

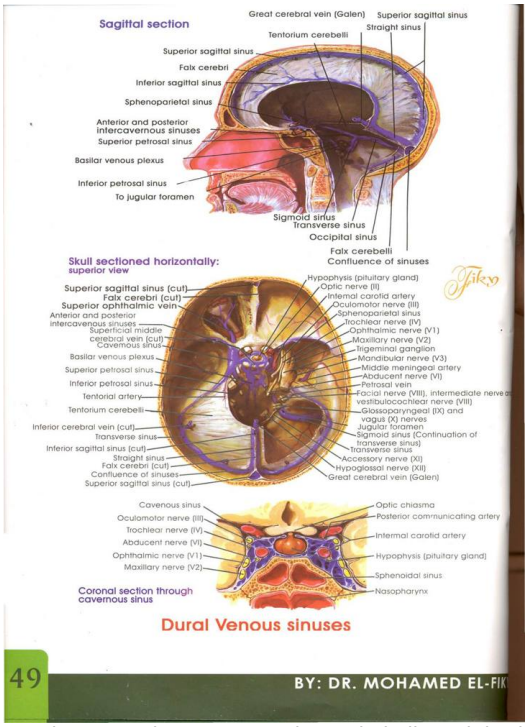


Foramina of the cranial base, superior view. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 6

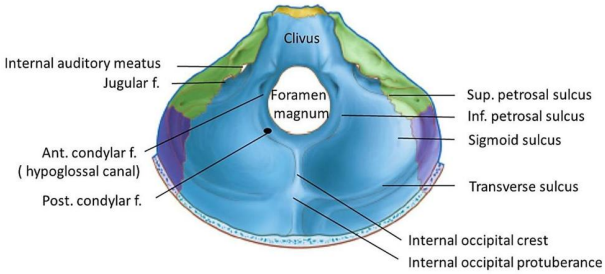
Posterior cranial fossa

- **Bones:** occipital, the back of the petrous and mastoid parts of the temporal, and a small part of the sphenoid (dorsum sellae and the clivus above the basi-occiput).
- **Internal acoustic meatus:** VII, VIII and the labyrinthine artery.
- **Jugular foramen:** IX, X, XI, the internal jugular vein (from the sigmoid sinus) and the inferior petrosal sinus.
- **Hypoglossal canal:** XII. **Foramen magnum:** medulla and meninges, vertebral arteries, spinal arteries and the spinal roots of XI.
- **Grooves:** transverse sinus, sigmoid sinus, inferior petrosal sinus.
- The full foramina table is under **Temporal bone & skull base**.

Clinical pearl: A jugular foramen lesion affects IX, X and XI together (Vernet's syndrome).



Dural venous sinuses. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 49

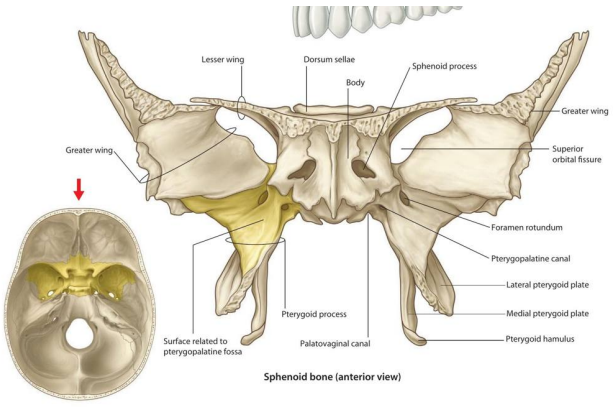


Posterior cranial fossa. Dr. Sameh Doss handout, p. 28

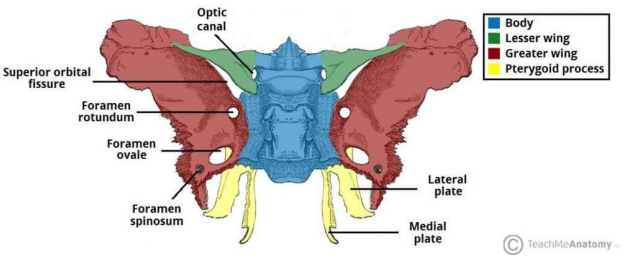
Sphenoid bone

- **Parts:** body (contains the sphenoidal sinus), two lesser wings, two greater wings, two pterygoid processes.
- **Greater wing foramina, front to back:** rotundum, ovale, spinosum.
- **Superior orbital fissure:** the gap between the lesser and greater wings. **Optic canal:** at the root of the lesser wing.
- **Pterygoid process:** medial plate ending in the hamulus, lateral plate, pterygoid fossa between them, scaphoid fossa at the top of the medial plate.
- **Pterygoid canal:** carries the nerve of the pterygoid canal (Vidian nerve) forward to the pterygopalatine fossa.

Clinical pearl: The tendon of tensor veli palatini turns around the hamulus to reach the palate.



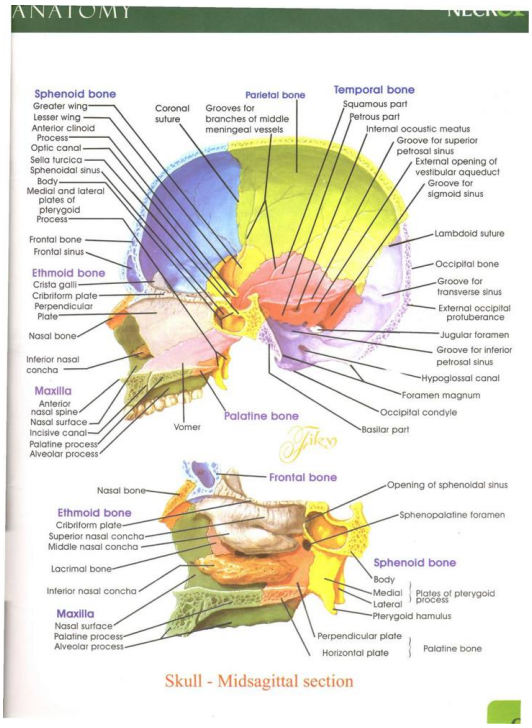
Sphenoid bone (anterior view). Gray's Atlas of Anatomy, 2nd ed., p. 534



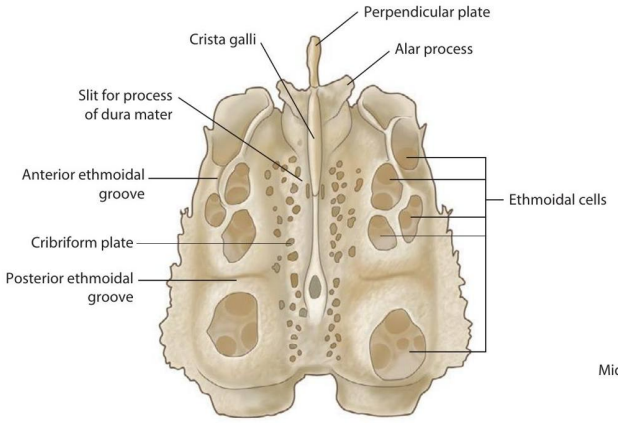
Sphenoid bone: body, wings and pterygoid processes. ENT Steps Anatomy, p. 147

Ethmoid, maxilla, palatine and small facial bones

- **Ethmoid:** cribriform plate with crista galli, perpendicular plate, and two labyrinths carrying the air cells, superior and middle conchae and uncinate process.
- **Maxilla:** a body containing the maxillary sinus, and **four processes:** frontal, zygomatic, alveolar and palatine.
- **Palatine bone:** L-shaped, with horizontal and perpendicular plates, a pyramidal process, and orbital and sphenoidal processes. The **sphenopalatine notch** between these last two forms the sphenopalatine foramen.
- **Nasal septum (bony part):** perpendicular plate of the ethmoid and the vomer.
- **Inferior concha** is a separate bone. The superior and middle conchae belong to the ethmoid.



Skull, midsagittal section. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 8



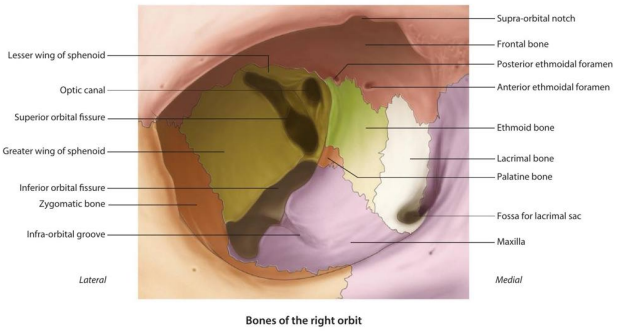
Ethmoid bone (superior view)

Ethmoid bone (superior view): crista galli and cribriform plate. Gray's Atlas of Anatomy, 2nd ed., p. 486

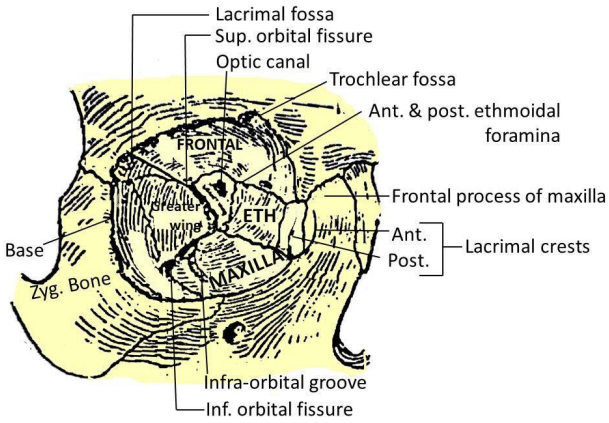
Bony orbit

- **Roof:** orbital plate of the frontal, lesser wing of sphenoid.
- **Medial wall:** frontal process of maxilla, lacrimal, orbital plate of ethmoid (lamina papyracea), body of sphenoid.
- **Floor:** maxilla, zygomatic and orbital process of the palatine.
- **Lateral wall:** zygomatic and greater wing of sphenoid.
- Fissures, relations and sinus surgery risks are covered under **Orbit & sinus relations**.

Clinical pearl: The thin lamina papyracea and the floor over the infra-orbital canal are the usual sites of blow-out fracture.



Bones of the right orbit. Gray's Atlas of Anatomy, 2nd ed., p. 508

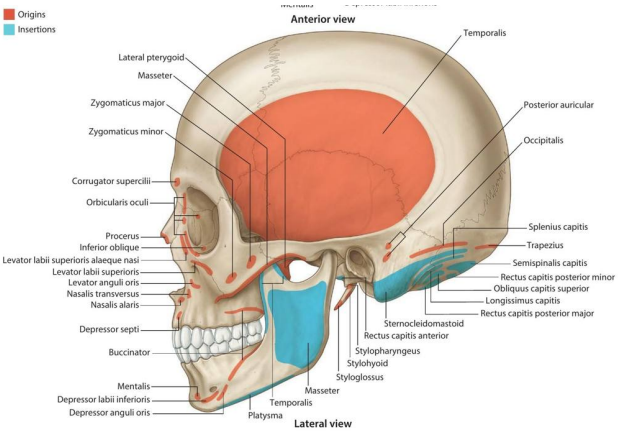


Bony orbit: walls, fissures and foramina. Dr. Sameh Doss handout, p. 90

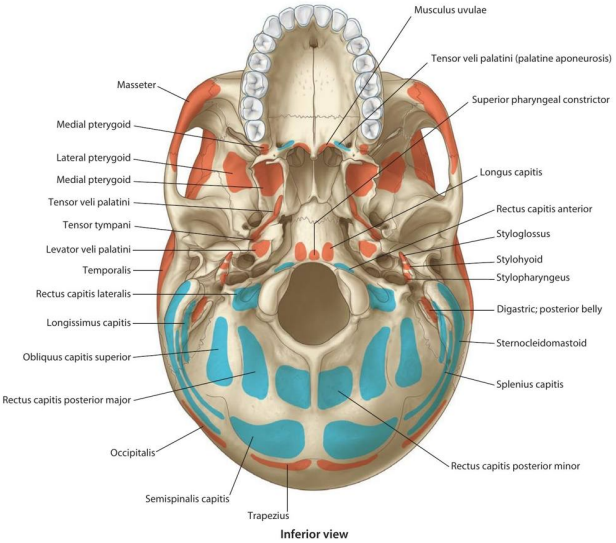
Muscle attachments on the skull

- **Temporal fossa:** temporalis. **Zygomatic arch:** masseter.
- **Mastoid process:** sternocleidomastoid, splenius capitis, longissimus capitis. **Mastoid notch:** posterior belly of digastric.
- **Styloid process:** styloglossus, stylohyoid, stylopharyngeus.
- **Pterygoid plates:** lateral pterygoid (lateral surface of the lateral plate), medial pterygoid (medial surface of the lateral plate).
- **Pharyngeal tubercle:** pharyngeal raphe. **Superior nuchal line:** trapezius and sternocleidomastoid.

Clinical pearl: Three styloid muscles have three different nerves: styloglossus XII, stylohyoid VII, stylopharyngeus IX.



Skull, lateral view: muscle attachments. Gray's Atlas of Anatomy, 2nd ed., p. 488

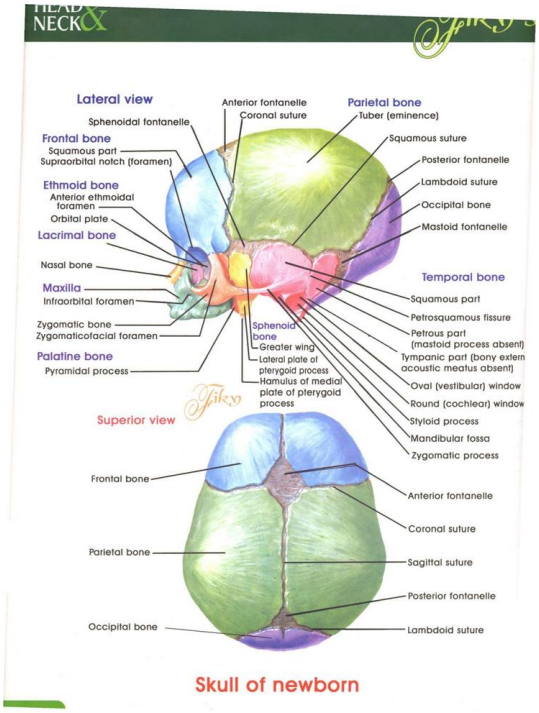


Skull, inferior view: muscle attachments. Gray's Atlas of Anatomy, 2nd ed., p. 489

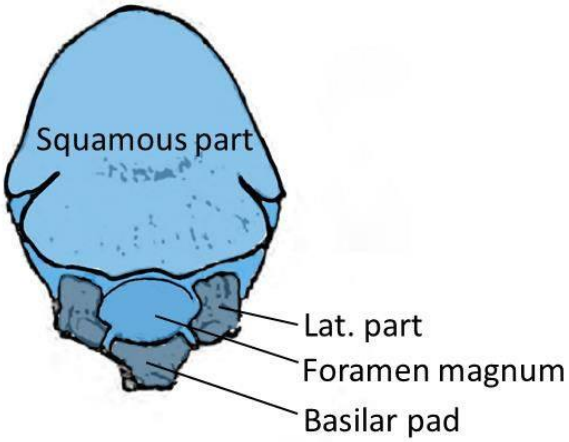
Skull of the newborn

- **Six fontanelles:** anterior, posterior, and paired sphenoidal and mastoid.
- **Anterior fontanelle:** where the coronal, sagittal and frontal sutures meet; it usually closes by about 18 months. **Posterior fontanelle:** where the sagittal and lambdoid sutures meet; it usually closes within the first few months.
- **Temporal bone at birth:** no mastoid process and no bony external acoustic meatus.

Clinical pearl: With no mastoid process, the facial nerve lies close to the surface in an infant, so a postauricular incision must be placed with care.



Skull of the newborn. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 9



Skull at birth: fontanelles. Dr. Sameh Doss handout, p. 37

Smaller openings of the cranial base

- **Foramen caecum:** emissary vein to the superior sagittal sinus.
- **Foramen of Vesalius (inconstant):** sphenoidal emissary vein, near the foramen ovale.
- **Hiatuses on the petrous bone:** greater and lesser petrosal nerves.
- **External opening of the vestibular aqueduct:** endolymphatic duct.
- **Mastoid foramen and condylar canal (both inconstant):** emissary veins.

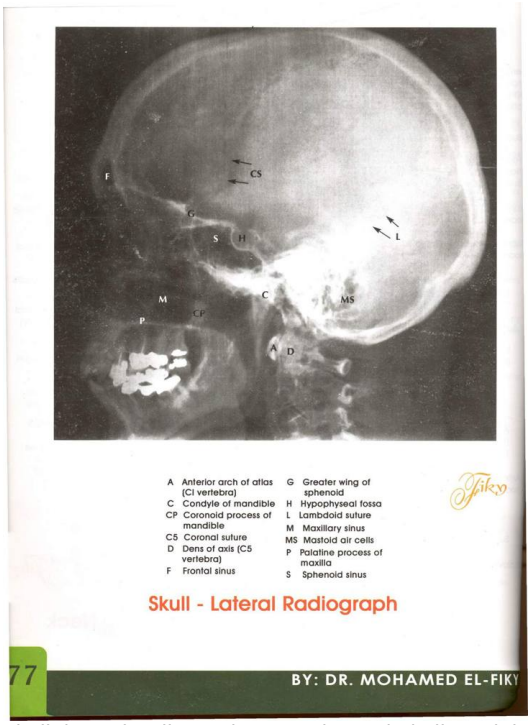
Meninges and dural venous sinuses

- **Dura:** covers the brain and folds to form the falx cerebri and tentorium cerebelli. The **middle meningeal artery** runs in the extradural plane.
- **Drainage pathway:** superior sagittal sinus → confluence → transverse → sigmoid → internal jugular vein at the jugular foramen.
- **Cavernous sinus:** drains through the superior and inferior petrosal sinuses.
- **Diploic and emissary veins** link the scalp, skull bones and dural sinuses.

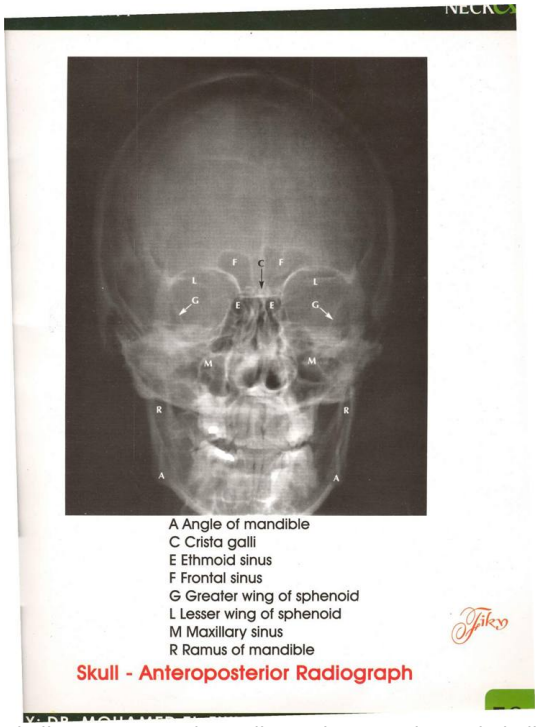
Clinical pearl: Emissary veins are a route for scalp or sinus infection to reach the dural sinuses.

On imaging

- **Lateral radiograph:** hypophyseal fossa, sphenoidal and frontal sinuses, mastoid air cells, coronal and lambdoid sutures, anterior arch of the atlas and the dens.
- **AP radiograph:** crista galli, frontal, ethmoidal and maxillary sinuses, greater and lesser wings of sphenoid, ramus and angle of the mandible.



Skull, lateral radiograph. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 77



Skull, anteroposterior radiograph. Dr. Mohamed El-Fiky, Mini Atlas of Anatomy Part V, p. 78

Skull base foramina and their contents

Foramen	Bone / fossa	Contents
Cribriform plate	Ethmoid · anterior fossa	Olfactory nerves (I), anterior ethmoidal nerve and vessels
Optic canal	Lesser wing of sphenoid	Optic nerve (II), ophthalmic artery
Superior orbital fissure	Between greater and lesser wings	III, IV, VI, V1 (lacrimal, frontal, nasociliary), superior ophthalmic vein
Foramen rotundum	Greater wing · middle fossa	Maxillary nerve (V2)
Foramen ovale	Greater wing · middle fossa	V3, accessory meningeal artery, lesser petrosal nerve, emissary vein (MALE)
Foramen spinosum	Greater wing · middle fossa	Middle meningeal artery and vein, meningeal branch of V3
Foramen lacerum	Between sphenoid, petrous and occipital	Cartilage; ICA crosses its upper part; greater + deep petrosal form the vidian nerve
Internal acoustic meatus	Petrous · posterior fossa	VII, nervus intermedius, VIII, labyrinthine artery and vein
Jugular foramen	Petrous + occipital · posterior fossa	IX, X, XI, IJV (sigmoid sinus), inferior petrosal sinus
Hypoglossal canal	Occipital	XII, meningeal branch of ascending pharyngeal artery
Stylomastoid foramen	Temporal	Facial nerve (VII), stylomastoid artery
Petrotympanic fissure	Temporal	Chorda tympani, anterior tympanic artery
Foramen magnum	Occipital	Medulla, vertebral arteries, spinal roots of XI
Foramen of Vesalius (inconstant)	Greater wing · middle fossa	Sphenoidal emissary vein
Hiatus for the greater petrosal nerve	Petrous · middle fossa	Greater petrosal nerve
Hiatus for the lesser petrosal nerve	Petrous · middle fossa	Lesser petrosal nerve
External opening of the vestibular aqueduct	Petrous · posterior fossa	Endolymphatic duct
Mastoid foramen (inconstant)	Temporal · posterior fossa	Emissary vein, occasional branch of occipital artery
Condylar canal (inconstant)	Occipital · posterior fossa	Emissary vein, meningeal branch of ascending pharyngeal artery

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